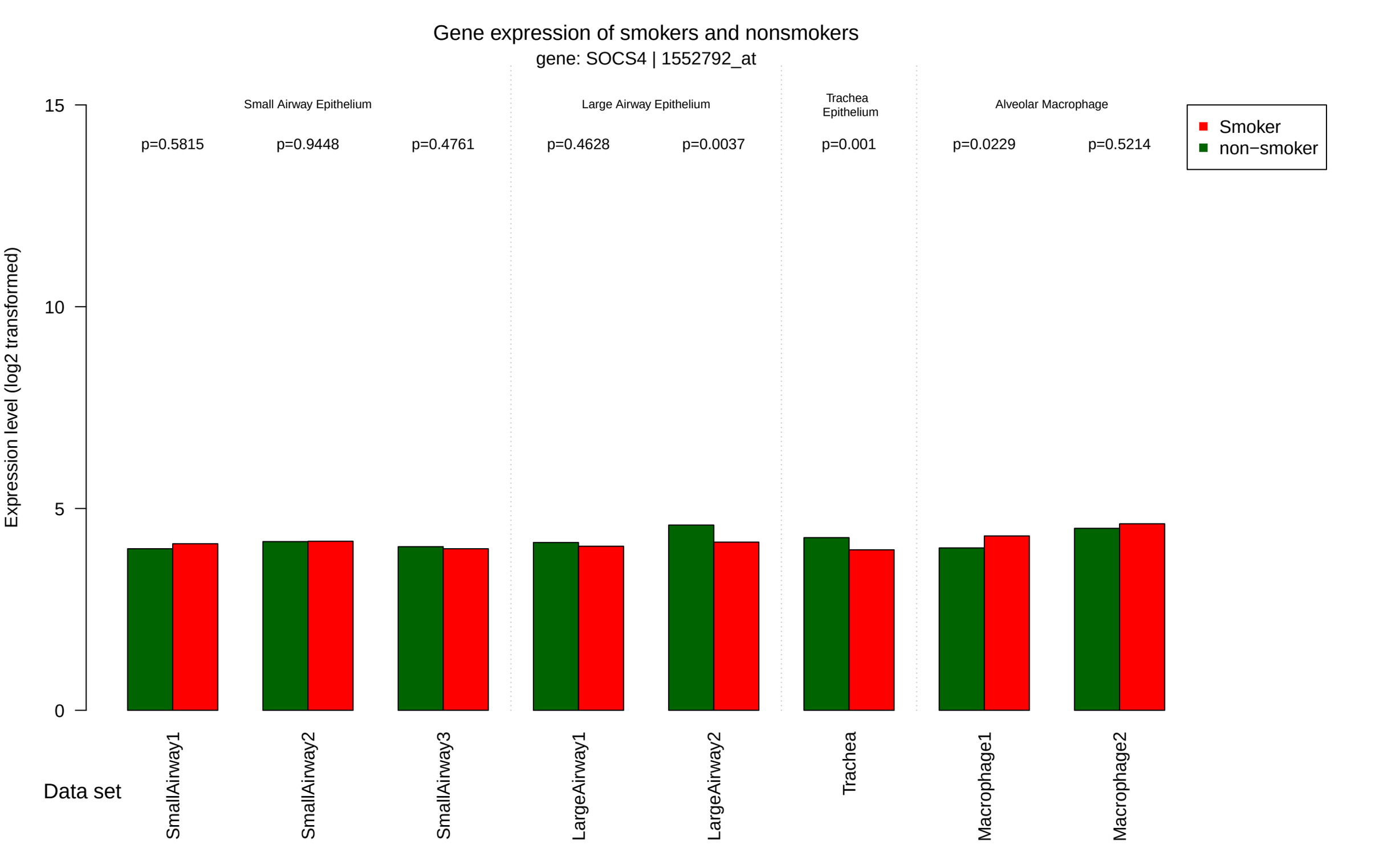
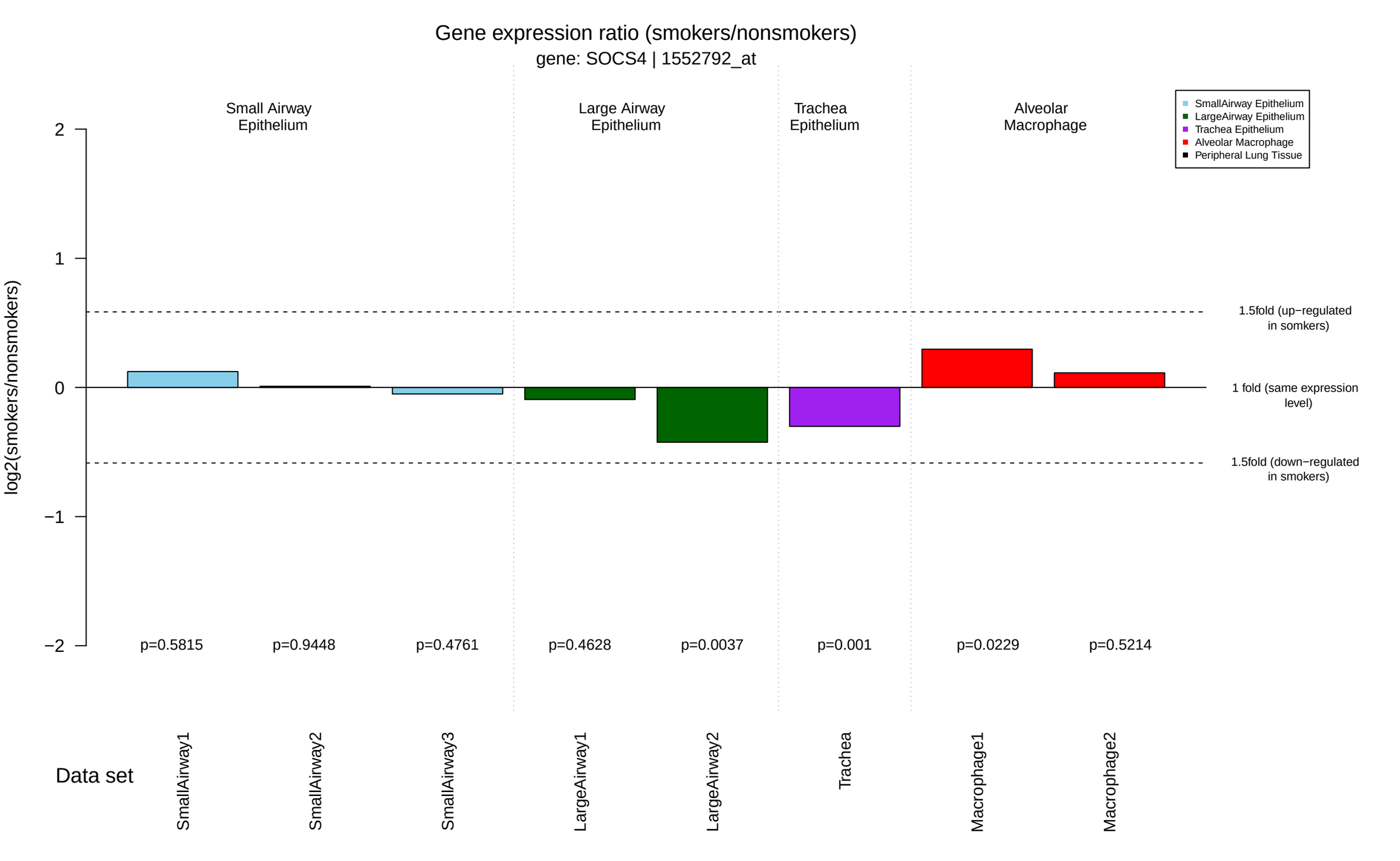
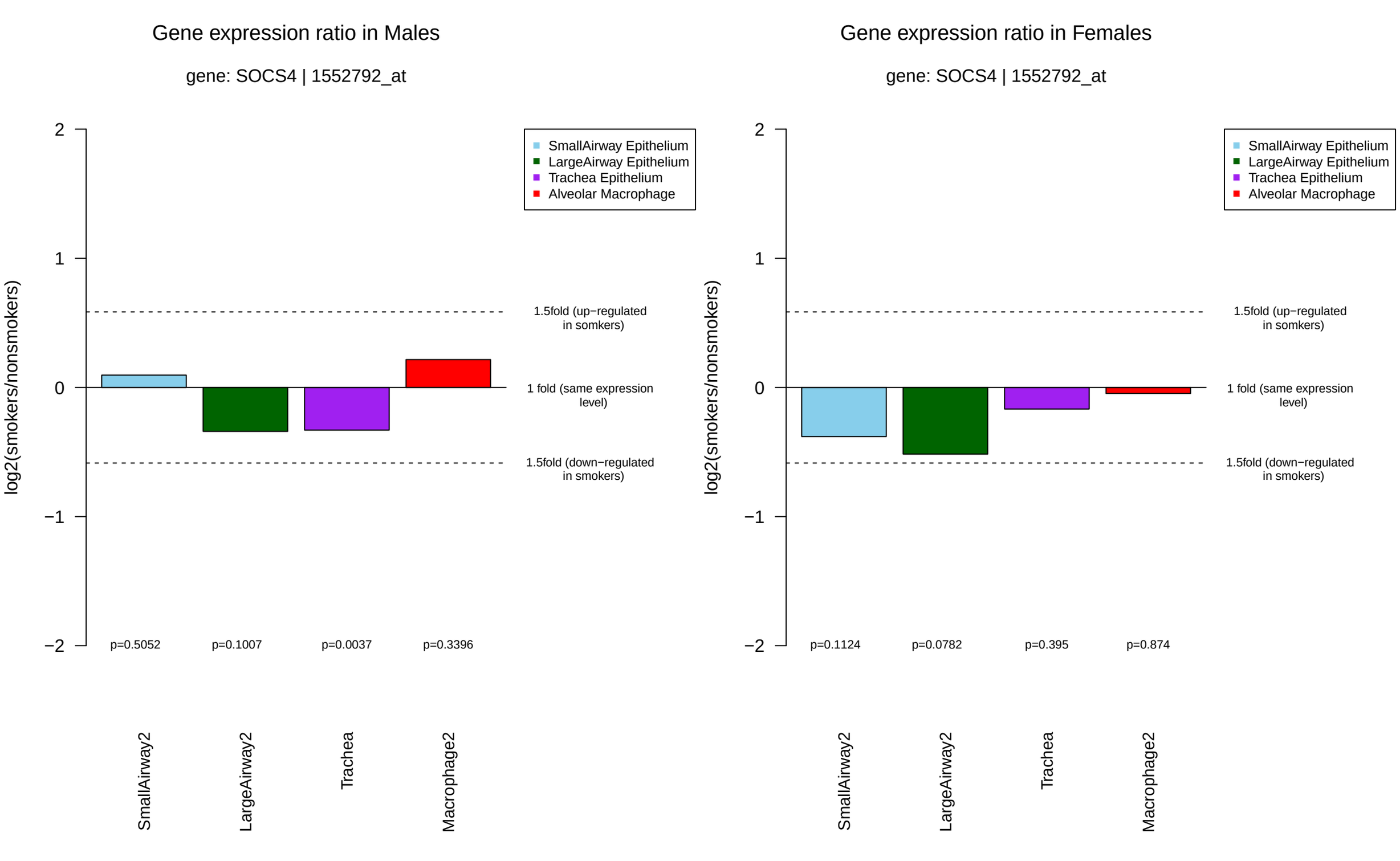
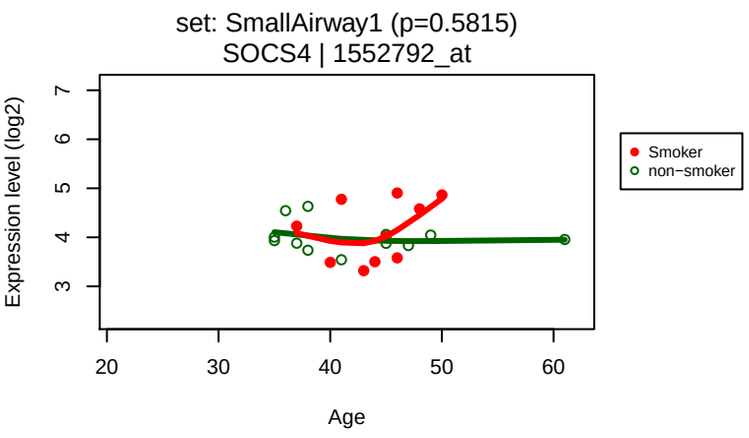
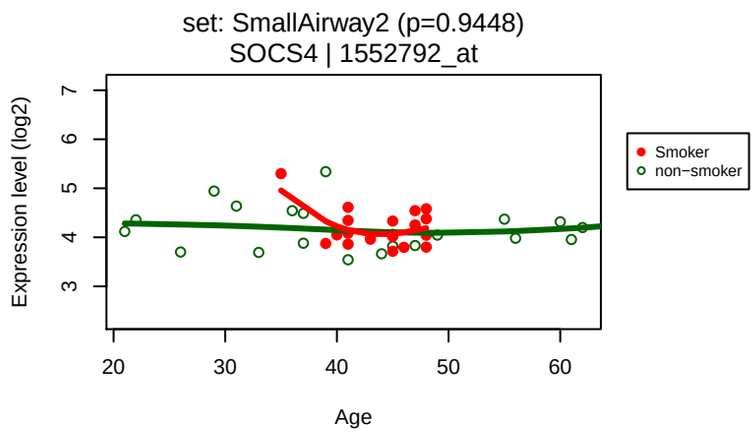
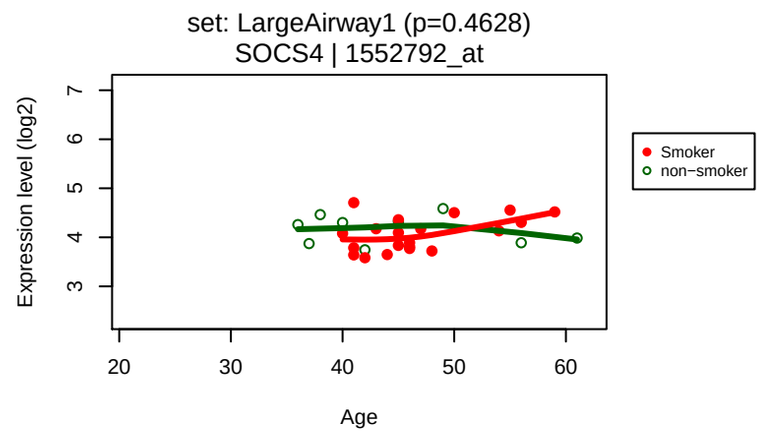

Gene expression of smokers and nonsmokers
gene: SOCS4 | 1552792_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5815
p=0.9448
p=0.4761
p=0.4628
p=0.0037
p=0.001
p=0.0229
p=0.5214
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

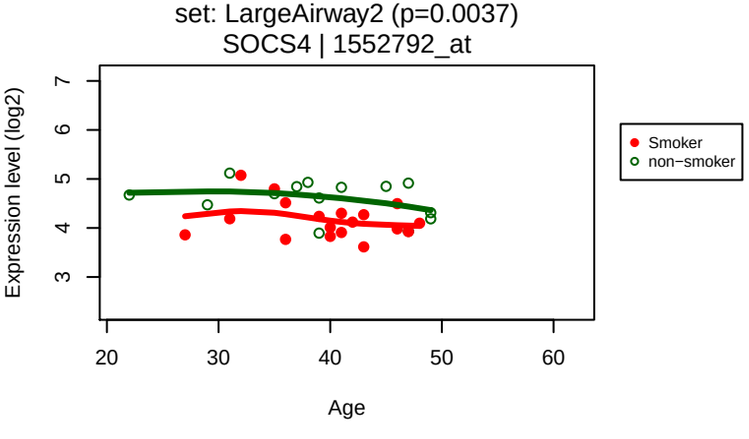

Gene expression ratio (smokers/nonsmokers)
gene: SOCS4 | 1552792_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5815
p=0.9448
p=0.4761
p=0.4628
p=0.0037
p=0.001
p=0.0229
p=0.5214
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

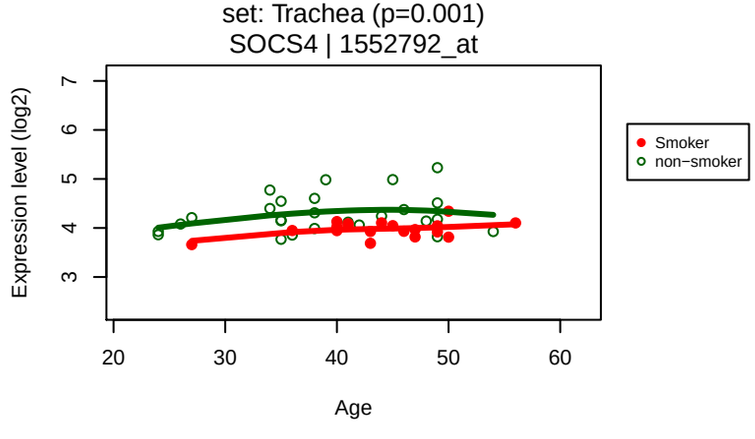

Gene expression ratio in Males
gene: SOCS4 | 1552792_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5052
p=0.1007
p=0.0037
p=0.3396
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOCS4 | 1552792_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1124
p=0.0782
p=0.395
p=0.874
SmallAirway2
LargeAirway2
Trachea
Macrophage2

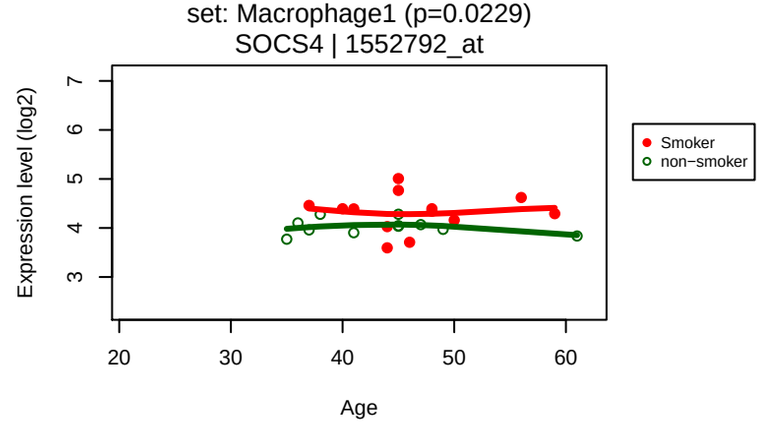

set: SmallAirway1 (p=0.5815)
SOCS4 | 1552792_at
Expression level (log2)
Age
Smoker
non−smoker

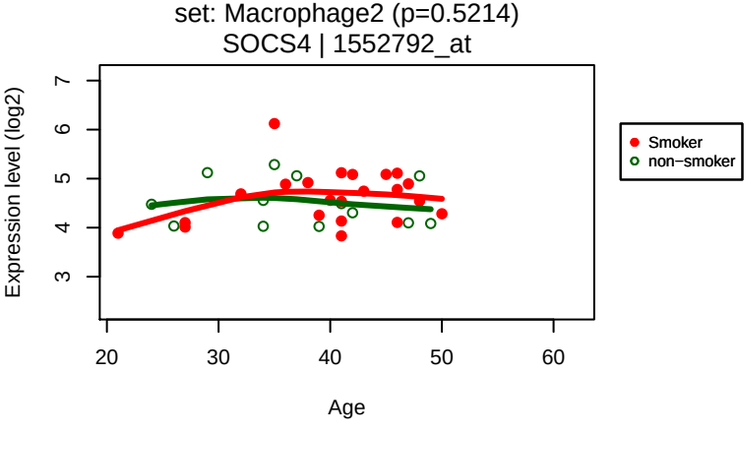

set: SmallAirway2 (p=0.9448)
SOCS4 | 1552792_at
Expression level (log2)
Age
Smoker
non−smoker

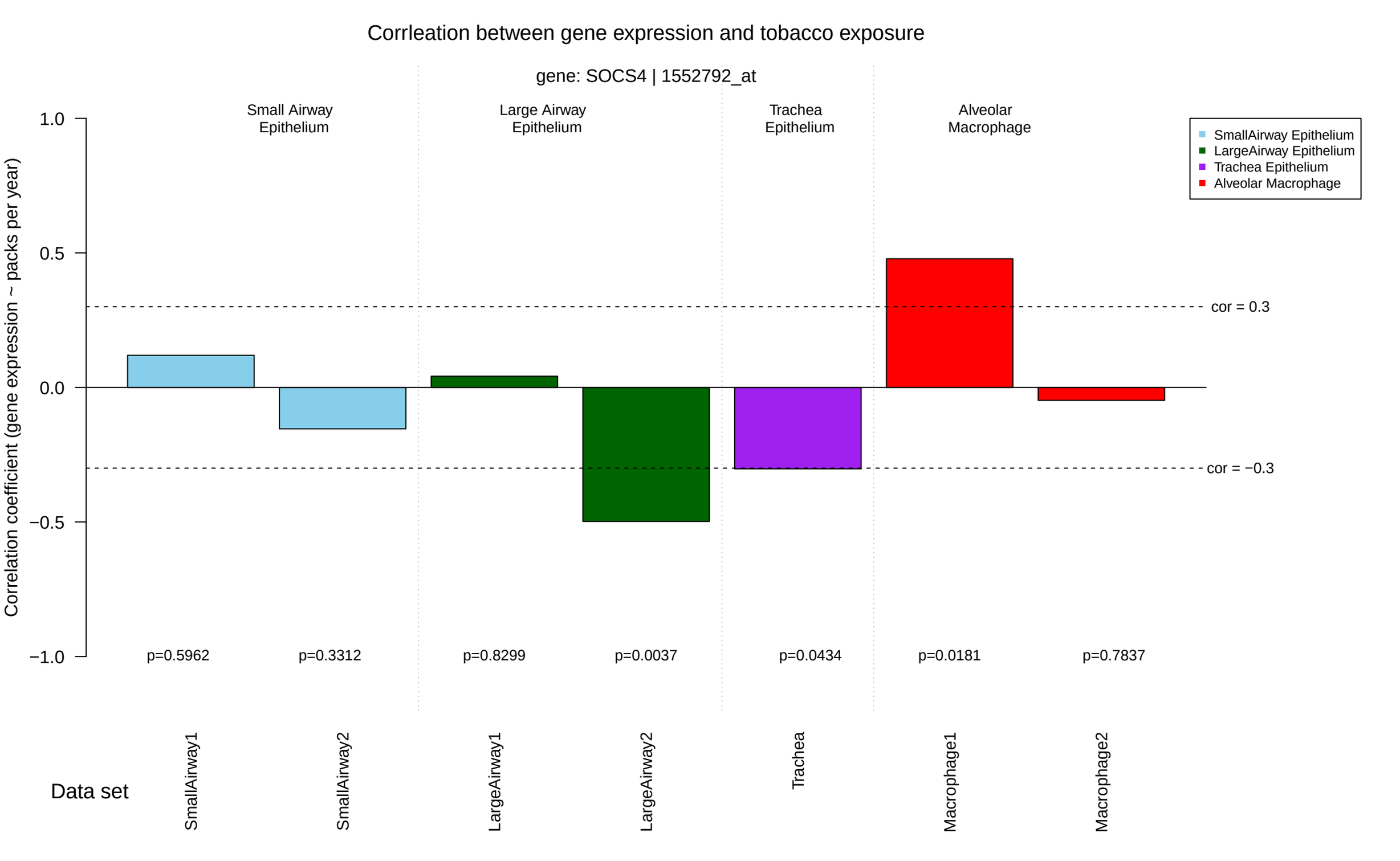

set: LargeAirway1 (p=0.4628)
SOCS4 | 1552792_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0037)
SOCS4 | 1552792_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.001)
SOCS4 | 1552792_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0229)
SOCS4 | 1552792_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5214)
SOCS4 | 1552792_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOCS4 | 1552792_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5962
p=0.3312
p=0.8299
p=0.0037
p=0.0434
p=0.0181
p=0.7837
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2